EVERY INVARIANT ELEMENT OF A QUOTIENT LIFTS TO A MONIC INVARIANT POLYNOMIAL MAPPING TO ITS NTH POWER

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Action of a finite group on a ring map and the invariant subrings A^G and B^G. the hypotheses at statement lines 32541-32542. Makes sense of the requirement that P lie in A^G[T] and that its image lie in B^G[T].
- Multiplicativity of a ring map on products and the fact that an invariant element is fixed by every group element. used at proof line 32549. Is what lets the image of prod (T - sigma(a)) be identified with prod (T - b) = (T - b)^n.

Lemma 0.2. *Let $\varphi : A \to B$ be a surjection of rings. Let G be a finite group of order n acting on $\varphi : A \to B$. If $b \in B^G$, then there exists a monic polynomial $P \in A^G[T]$ which maps to $(T - b)^n$ in $B^G[T]$.*

Proof. Choose $a \in A$ lifting b and set $P = \prod_{\sigma \in G}(T - \sigma(a))$. □

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